

ECE 222 Homework #1

Due Thursday, September 10, 2026 at 11:59 pm on Canvas

Scan or photograph and save as a single .pdf file.

Name your file "HW_No_1_Last Name.pdf"

Do problems 3.36, 3.37, 3.42, 3.44, 3.58

$$\therefore \text{grad} = \nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle$$

3.36 Find the gradient of the following scalar functions:

- a. $T = 3/(x^2 + z^2)$
- b. $V = xy^2z^4$
- c. $U = z \cos \phi / (1 + r^2)$
- d. $W = e^{-R} \sin \theta$
- *e. $S = 4x^2e^{-z} + y^3$
- f. $N = r^2 \cos^2 \phi$
- g. $M = R \cos \theta \sin \phi$

See Page 8 of this Pdf for my Answers

3.37 For each of the following scalar fields, obtain an analytical solution for ∇T and generate a corresponding arrow representation.

- a. $T = 10 + x$, for $-10 \leq x \leq 10$
- *b. $T = x^2$, for $-10 \leq x \leq 10$
- b. $T = 100 + xy$, for $-10 \leq x \leq 10$
- c. $T = x^2y^2$, for $-10 \leq x, y \leq 10$
- e. $T = 20 + x + y$, for $-10 \leq x, y \leq 10$
- f. $T = 1 + \sin(\pi x/3)$, for $-10 \leq x \leq 10$
- *g. $T = 1 + \cos(\pi x/3)$, for $-10 \leq x \leq 10$
- h. $T = 15 + r \cos \phi$, for $\begin{cases} 0 \leq r \leq 10 \\ 0 \leq \phi \leq 2\pi \end{cases}$
- i. $T = 15 + r \cos^2 \phi$, for $\begin{cases} 0 \leq r \leq 10 \\ 0 \leq \phi \leq 2\pi \end{cases}$

See Pages 8 & 9

NOTE: See the figures in problem 3.44 for examples of "arrow representations". You can make approximate sketches by hand.

3.42 For the scalar function $T = \frac{1}{2} e^{-r/5} \cos \phi$, determine its directional derivative along the radial direction $\hat{\mathbf{r}}$ and then evaluate it at $P = (2, \pi/4, 3)$.

$D_{\hat{\mathbf{r}}} T = \nabla T \cdot \frac{\mathbf{r}}{\|\mathbf{r}\|}; D_{\hat{\mathbf{r}}} T = \frac{\partial T}{\partial r}$

$\nabla T = \left\langle -\frac{1}{10} e^{-r/5} \cos \phi, -\frac{1}{20} e^{-r/5} \sin \phi, 0 \right\rangle$

$D_{\hat{\mathbf{r}}} T = -\frac{1}{10} e^{-r/5} \cos \phi; D_{\hat{\mathbf{r}}} T(2, \pi/4, 3) = -\frac{1}{10} e^{-2/5} \cos(\pi/4)$

$= -\frac{\sqrt{2}}{20} e^{-2/5} = -0.0474$

3.44 Each of the following vector fields is displayed in Fig. P3.44 in the form of a vector representation. Determine $\nabla \cdot \mathbf{A}$ analytically and then compare the result with your expectations on the basis of the displayed pattern.

a. $\mathbf{A} = -\hat{x} \cos x \sin y + \hat{y} \sin x \cos y$, for $-\pi \leq x, y \leq \pi$

Figure P3.44(a)

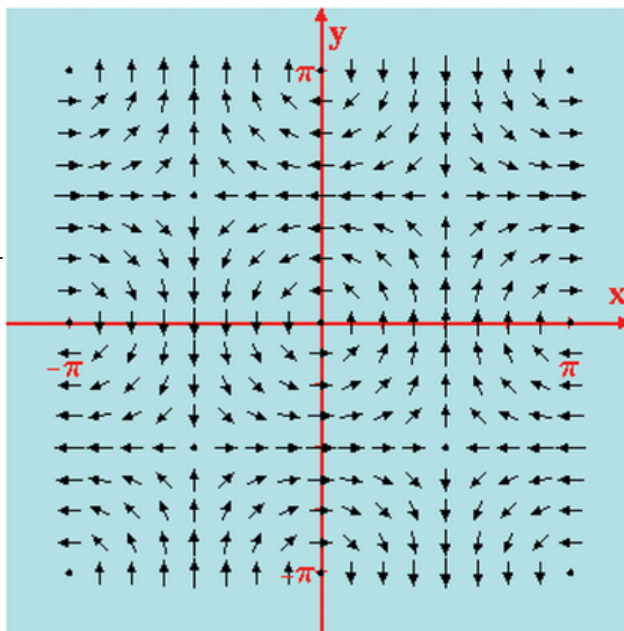
$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} = -\sin x \sin y - \sin x \sin y = -2 \sin x \sin y$$

$$\nabla \times \mathbf{A} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

$$(\cos x \cos y - (-\cos x \cos y)) = 2 \cos x \cos y$$

So: Solenoidal & not conservative

Curl +, - When $2 \cos x \cos y$ is + and - respectively; signs flip @ $x, y = \pm \frac{\pi}{2}$



By looking at it, there is curl at (0,0) about z; it looks like there is no div! = 0 anywhere...

I was

Right

b. $\mathbf{A} = -\hat{x} \sin 2y + \hat{y} \cos 2x$, for $-\pi \leq x, y \leq \pi$

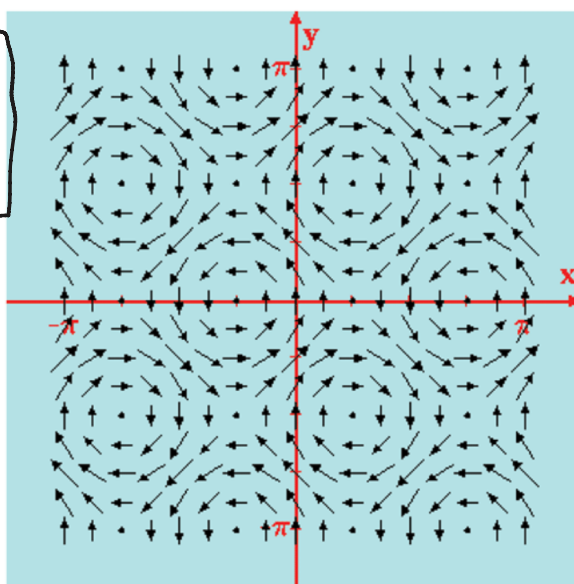
Figure P3.44(b)

$$\nabla \cdot \mathbf{A} = 0 + 0 = 0$$

$$\nabla \times \mathbf{A} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

$$-2 \sin 2x + 2 \cos 2y$$

Solenoidal, not conservative



div looks like zero everywhere...

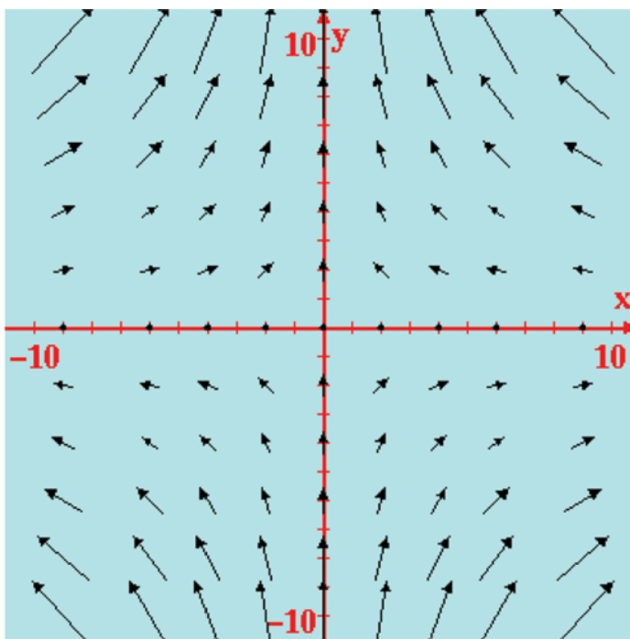
I was

Right

c. $\mathbf{A} = -\hat{x} xy + \hat{y} y^2$, for $-10 \leq x, y \leq 10$

Figure P3.44(c)

$$\nabla \cdot \mathbf{A} = -y + 2y = y$$



Looks somewhat divergent;

$$\nabla \cdot \mathbf{A} > 0 \text{ if } y > 0$$

$$\nabla \cdot \mathbf{A} < 0 \text{ if } y < 0$$

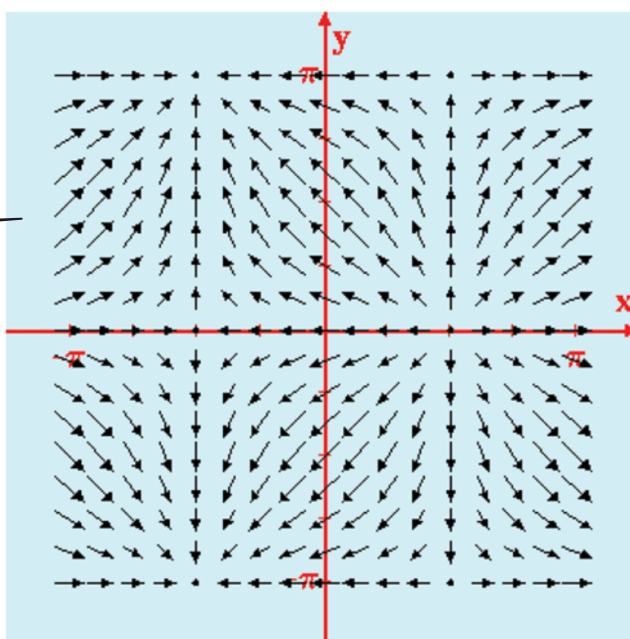
I was
Right

d. $\mathbf{A} = -\hat{x} \cos x + \hat{y} \sin y$, for $-\pi \leq x, y \leq \pi$

Figure P3.44(d)

$$\nabla \cdot \mathbf{A} = \sin x + \cos y, \quad \nabla \times \mathbf{A} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Not Solenoidal & Conservative



$\text{Div} > 0$ @ (2,0)
curl looks like zero
everywhere

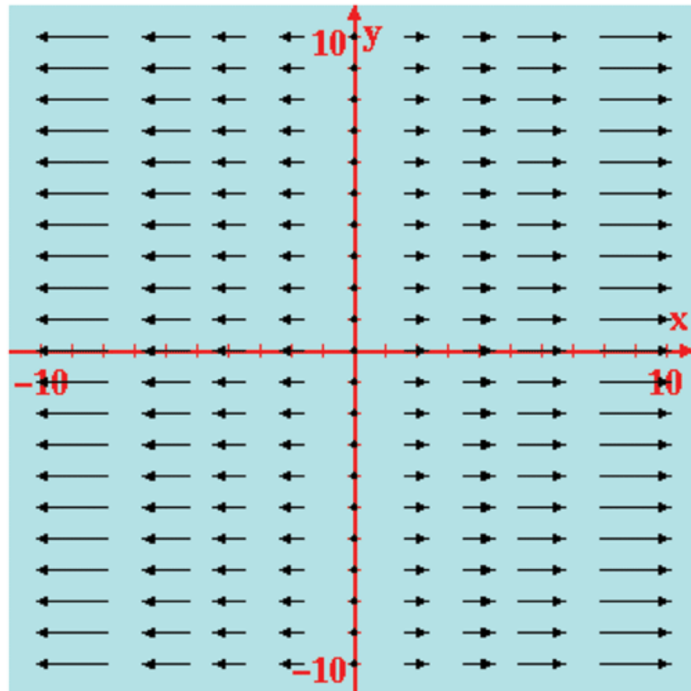
I was
Right

e. $\mathbf{A} = \hat{\mathbf{x}} x$, for $-10 \leq x \leq 10$

Figure P3.44(e)

$$\nabla \cdot \mathbf{A} = 1; \nabla \times \mathbf{A} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Not Solenoidal but Conservative



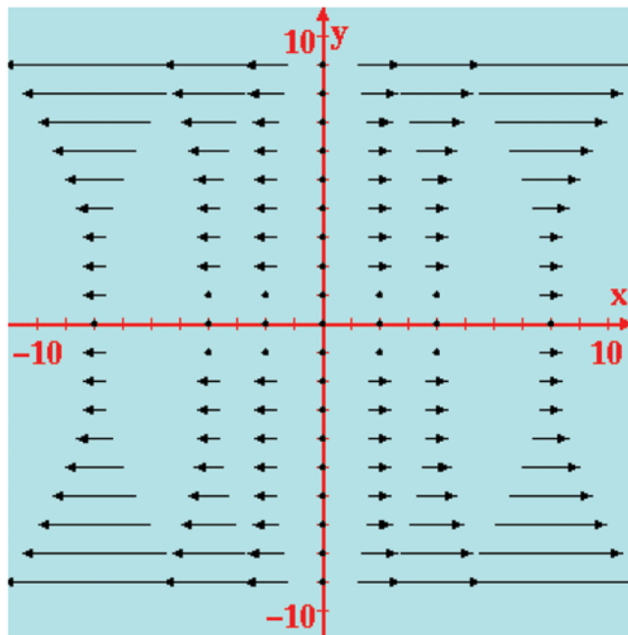
Maybe $\text{div} > 0$?
Curl is zero everywhere
it looks.

I was
Right

f. $\mathbf{A} = \hat{\mathbf{x}} xy^2$, for $-10 \leq x, y \leq 10$

Figure P3.44(f)

$$\nabla \cdot \mathbf{A} = y^2$$



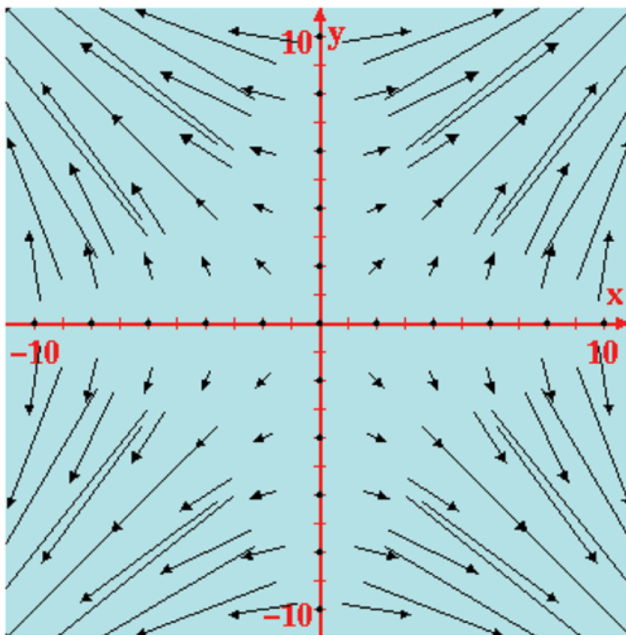
Looks
divergent
 $\nabla \cdot \mathbf{A} \geq 0$

I was
Right

g. $\mathbf{A} = \hat{x} xy^2 + \hat{y} x^2 y$, for $-10 \leq x, y \leq 10$

Figure P3.44(g)

$\nabla \cdot \mathbf{A} = y^2 + x^2$, $\nabla \times \mathbf{A} = \begin{bmatrix} 0-0 \\ 0-0 \\ 2xy-2xy=0 \end{bmatrix}$
 Not Solenoidal and
 Conservative



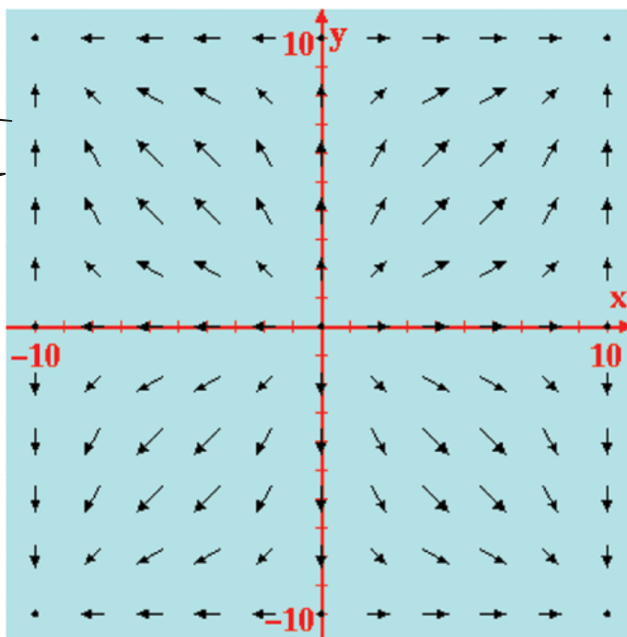
$\text{div} > 0$ @ (3,3)
 it seems
 $\text{Curl} = 0$ everywhere
 I think

I was
 Right

h. $\mathbf{A} = \hat{x} \sin\left(\frac{\pi x}{10}\right) + \hat{y} \sin\left(\frac{\pi y}{10}\right)$, for $-10 \leq x, y \leq 10$

Figure P3.44(h)

$\nabla \cdot \mathbf{A} = \frac{\pi}{10} \cos\left(\frac{\pi x}{10}\right) + \frac{\pi}{10} \cos\left(\frac{\pi y}{10}\right)$
 $\nabla \times \mathbf{A} = \begin{bmatrix} 0-0 \\ 0-0 \\ 0-0 \end{bmatrix}$
 Not Solenoidal & Conservative



$\text{div} > 0$ @ (3,3)
 Looks Conservative

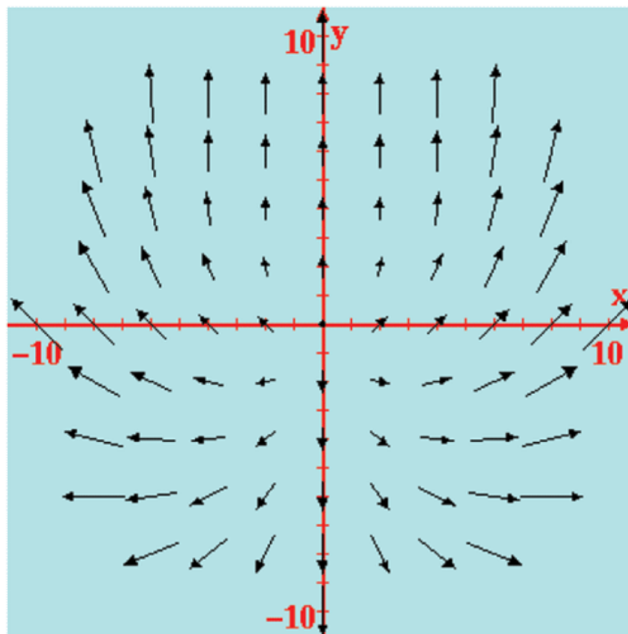
I was
 Right

i. $\mathbf{A} = \hat{\mathbf{r}} r + \hat{\boldsymbol{\phi}} r \cos \phi$, for $\begin{cases} 0 \leq r \leq 10 \\ 0 \leq \phi \leq 2\pi. \end{cases}$

Figure P3.44(i)

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial(r^2)}{\partial r} + \frac{1}{r} \frac{\partial(r \cos \phi)}{\partial \phi}$$

$$\nabla \cdot \mathbf{A} = 2 - \sin \phi$$



looks very
divergent
 $\nabla \cdot \mathbf{A} > 0$

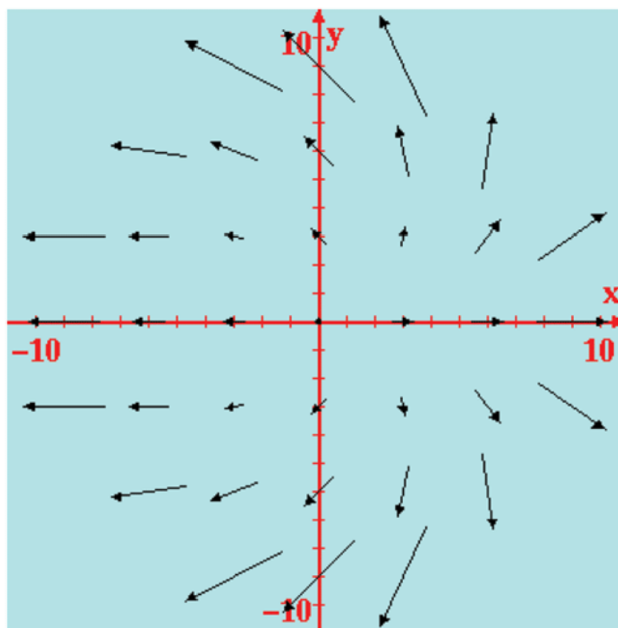
I was
Right

j. $\mathbf{A} = \hat{\mathbf{r}} r^2 + \hat{\boldsymbol{\phi}} r^2 \sin \phi$, for $\begin{cases} 0 \leq r \leq 10 \\ 0 \leq \phi \leq 2\pi. \end{cases}$

Figure P3.44(j)

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial(r^3)}{\partial r} + \frac{1}{r} \frac{\partial(r^2 \sin \phi)}{\partial \phi}$$

$$\nabla \cdot \mathbf{A} = r(3 + \cos \phi)$$



Looks divergent
 $\nabla \cdot \mathbf{A} > 0$

I was
Right

3.58 Determine if each of the following vector fields is solenoidal, conservative, or both:

* a. $\mathbf{A} = \hat{\mathbf{x}}x^2 - \hat{\mathbf{y}}2xy$

b. $\mathbf{B} = \hat{\mathbf{x}}x^2 - \hat{\mathbf{y}}y^2 + \hat{\mathbf{z}}2z$

c. $\mathbf{C} = \hat{\mathbf{r}}(\sin \phi)/r^2 + \hat{\phi}(\cos \phi)/r^2$

* d. $\mathbf{D} = \hat{\mathbf{R}}/R$

e. $\mathbf{E} = \hat{\mathbf{r}} \left(3 - \frac{r}{1+r} \right) + \hat{\mathbf{z}}z$

f. $\mathbf{F} = (\hat{\mathbf{x}}y + \hat{\mathbf{y}}x)/(x^2 + y^2)$

g. $\mathbf{G} = \hat{\mathbf{x}}(x^2 + z^2) - \hat{\mathbf{y}}(y^2 + x^2) - \hat{\mathbf{z}}(y^2 + z^2)$

* h. $\mathbf{H} = \hat{\mathbf{R}}(Re^{-R})$

See pages 18/16

Answers to starred problems:

3.36 $\square$ (e) $\nabla S = \hat{\mathbf{x}}8xe^{-z} + \hat{\mathbf{y}}3y^2 - \hat{\mathbf{z}}4x^2e^{-z}$

3.37 $\square$ (b) $\nabla T = \hat{\mathbf{x}}2x$

(g) $\nabla T = -\hat{\mathbf{x}} \frac{2\pi}{6} \sin\left(\frac{\pi x}{3}\right)$

3.58 $\square$ (a) $\mathbf{A}$ is solenoidal, but not conservative.

(d) $\mathbf{D}$ is conservative, but not solenoidal.

(h) $\mathbf{H}$ is conservative, but not solenoidal.

HW1 Adam Shepanic

3.36: $\therefore \nabla = \text{grad} = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle$

a. $T = 3/(x^2+z^2)$; $\frac{\partial}{\partial x} T = \frac{\partial}{\partial x} \left(\frac{3}{(x^2+z^2)} \right)$; $\frac{\partial}{\partial x} T = \frac{\partial}{\partial x} (3(x^2+z^2)^{-1})$

$\frac{\partial}{\partial x} T = -6x(x^2+z^2)^{-2}$; $\frac{\partial}{\partial z} T = -6z(x^2+z^2)^{-2}$; $\nabla T = \left\langle \frac{-6x}{(x^2+z^2)^2}, 0, \frac{-6z}{(x^2+z^2)^2} \right\rangle$

b. $\nabla V = \langle y^2 z^4, 2xy z^4, 4xy^2 z^3 \rangle$

c. $\therefore \nabla$, grad in cylindrical coords $\rightarrow \hat{r} \frac{\partial T}{\partial r} + \hat{\phi} \frac{1}{r} \frac{\partial T}{\partial \phi} + \hat{z} \frac{\partial T}{\partial z}$

$U = 2 \cos \phi / (1+r^2)$; $\frac{\partial}{\partial r} U = \frac{-2r \cos \phi}{(1+r^2)^2}$; $\frac{\partial}{\partial \phi} U = -2 \sin \phi / (1+r^2)$

$\frac{\partial}{\partial z} U = \cos \phi / (1+r^2)$; $\nabla U = \left\langle \frac{-2r \cos \phi}{(1+r^2)^2}, \frac{-2 \sin \phi}{r(1+r^2)}, \frac{\cos \phi}{(1+r^2)} \right\rangle$

d. $\therefore$ grad for Spherical $\rightarrow$

$\nabla W = \hat{r} \frac{\partial T}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial T}{\partial \theta} + \hat{\phi} \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi}$

$W = e^R \sin \theta$; $\frac{\partial}{\partial R} W = e^R \sin \theta$; $\frac{\partial}{\partial \theta} W = e^R \cos \theta$; $\frac{\partial}{\partial \phi} W = 0$

$\nabla W = \langle e^R \sin \theta, e^R \cos \theta, 0 \rangle$

e. $S = 4x^2 e^{-z} + y^3$; $\frac{\partial}{\partial x} S = 8x e^{-z}$; $\frac{\partial}{\partial y} S = 3y^2$; $\frac{\partial}{\partial z} S = -4x^2 e^{-z}$

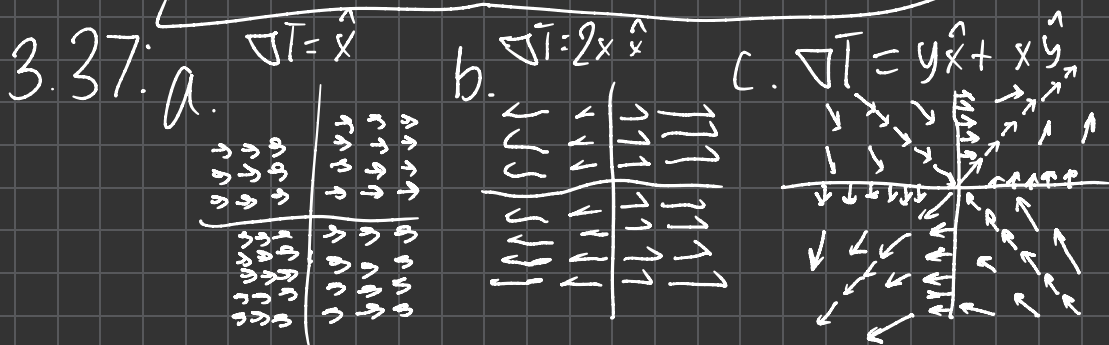
$\nabla S = \langle 8x e^{-z}, 3y^2, -4x^2 e^{-z} \rangle$

f. $N = r^2 \cos^2 \phi$; $\frac{\partial}{\partial r} N = 2r \cos^2 \phi$; $\frac{\partial}{\partial \phi} N = -2r^2 \sin \phi \cos \phi$; $\frac{\partial}{\partial z} = 0$

$\nabla N = \langle 2r \cos^2 \phi, -2r^2 \sin \phi \cos \phi, 0 \rangle$

g. $M = R \cos \theta \sin \phi$; $\frac{\partial}{\partial R} M = \cos \theta \sin \phi$; $\frac{\partial}{\partial \theta} M = -R \sin \theta \sin \phi$; $\frac{\partial}{\partial \phi} M = R \cos \theta \cos \phi$

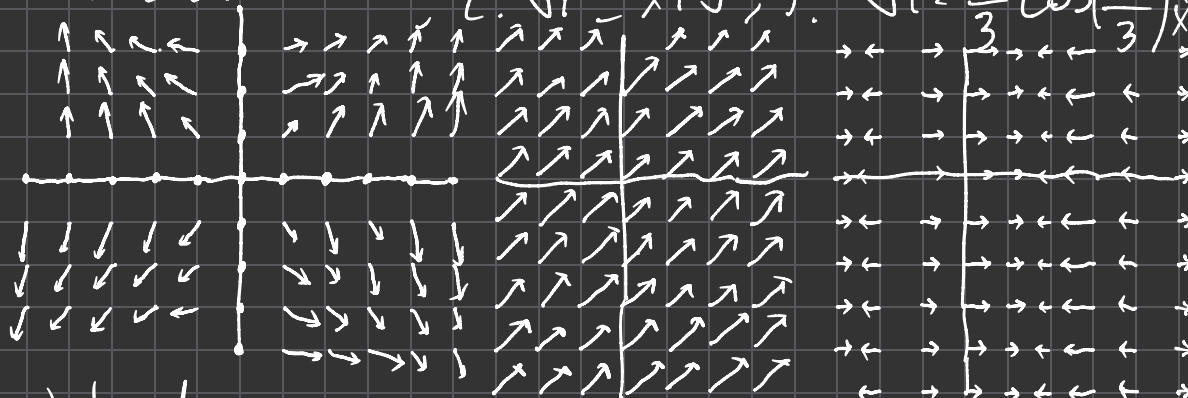
$\nabla M = \langle \cos \theta \sin \phi, -R \sin \theta \sin \phi, R \cos \theta \cos \phi \rangle$



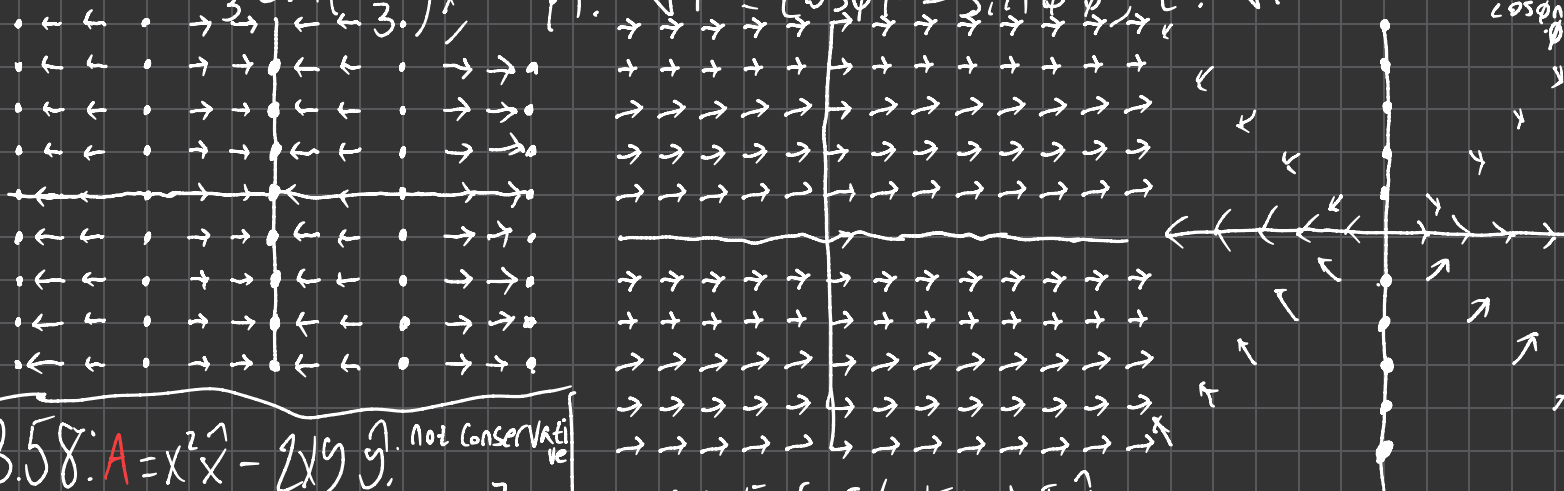


Continued:

3.37: d. $\nabla T = 2xy^2\hat{x} + 2x^2y\hat{y}$; e. $\nabla T = \hat{x} + \hat{y}$; f. $\nabla T = \frac{\pi}{3} \cos(\frac{\pi x}{3})\hat{x}$



g. $\nabla T = -\frac{\pi}{3} \sin(\frac{\pi x}{3})\hat{x}$; h. $\nabla T = \cos\phi\hat{r} - \sin\phi\hat{\phi}$; i. $\nabla T = \cos^2\phi\hat{r} - 2\sin\phi\cos\phi\hat{\phi}$



3.58: $A = x^2\hat{x} - 2xy\hat{y}$. Not Conservative

$\nabla \cdot A = 2x - 2x = 0$; Solenoidal; $\nabla \times A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ -2y & 0 \end{bmatrix}$

Cylindrical formulae

$\nabla \times A = \frac{1}{r} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z}$
 $\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}$
 $\frac{1}{r} \left(\frac{\partial (rA_\phi)}{\partial r} - \frac{\partial A_r}{\partial \phi} \right)$

Spherical formulae

$B = B_r\hat{r} + B_\theta\hat{\theta} + B_\phi\hat{\phi}$
 $\nabla \cdot B = \frac{1}{r^2} \frac{\partial (r^2 B_r)}{\partial r} + \frac{1}{r \sin\theta} \frac{\partial (\sin\theta B_\theta)}{\partial \theta} + \frac{1}{r \sin\theta} \frac{\partial B_\phi}{\partial \phi}$
 $\nabla \times B = \frac{1}{r \sin\theta} \left(\frac{\partial (\sin\theta B_\phi)}{\partial \theta} - \frac{\partial B_\theta}{\partial \phi} \right)$
 $\frac{1}{r} \left(\frac{\partial (rA_\phi)}{\partial r} - \frac{\partial A_r}{\partial \phi} \right)$

$\therefore \nabla \cdot A = \frac{1}{r} \frac{\partial (rA_r)}{\partial r} + \frac{1}{r} \frac{\partial A_\theta}{\partial \phi} + \frac{\partial A_z}{\partial z}$

$C = \left(\frac{\sin\phi}{r^2} \right) \hat{r} + \left(\frac{\cos\phi}{r^2} \right) \hat{\phi}$; $r \cdot C_\phi = \frac{\cos\phi}{r}$

$\nabla \cdot C = -\left(\frac{\sin\phi}{r^3} \right) - \frac{\sin\phi}{r^3} = -\frac{2\sin\phi}{r^3}$

$\nabla \times C = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \frac{1}{r} \left(-\frac{\cos\phi}{r^2} - \frac{\cos\phi}{r^2} \right) \end{bmatrix}$; $\therefore$ Neither

$\frac{1}{r} \left(-\frac{\cos\phi}{r^2} - \frac{\cos\phi}{r^2} \right) = -\frac{2\cos\phi}{r^3}$

$D = \hat{r}/R$; $\nabla \cdot D = \frac{1}{R} \left(\frac{\partial (R)}{\partial R} \right) = \frac{1}{R^2}$; $\nabla \times D = 0$
 $\therefore$ Conservative only

$E = \left(3 - \frac{r}{r+1} \right) \hat{r} + 2\hat{z}$; $\nabla \cdot E = \frac{1}{r} \left(\frac{\partial (2r + \frac{r}{r+1})}{\partial r} \right) + 1$; $\nabla \cdot E = \frac{1}{r} \left(2 + \frac{1}{(r+1)^2} \right) + 1$
 $\nabla \times E = 0$; Conservative only

$F = \frac{y}{(x^2+y^2)} \hat{x} + \frac{x}{(x^2+y^2)} \hat{y}$; $\nabla \cdot F = -\frac{2xy}{(x^2+y^2)^2} - \frac{2yx}{(x^2+y^2)^2} = -\frac{4xy}{(x^2+y^2)^2}$
 $\nabla \times F = \frac{(x^2+y^2) - 2x^2}{(x^2+y^2)^2} \hat{z} - \frac{x^2+y^2 - 2y^2}{(x^2+y^2)^2} \hat{z} = \frac{2(y^2-x^2)}{(x^2+y^2)^2} \hat{z}$; $\therefore$ Neither

$$3.58: \mathbf{G} = (x^2+z^2)\hat{x} - (x^2+y^2)\hat{y} - (y^2+z^2)\hat{z}; \nabla \cdot \mathbf{G} = 2x - 2y - 2z$$

$$\nabla \times \mathbf{G} = \begin{bmatrix} -2y \\ 2z \\ -2x \end{bmatrix} \therefore \text{Neither}$$

$$\mathbf{H} = R e^{-R} \hat{r}; \nabla \cdot \mathbf{H} = \frac{1}{R^2} \frac{\partial}{\partial R} (R^3 e^{-R}) = \frac{1}{R^2} (3R^2 e^{-R} - R^3 e^{-R}) = (3-R) e^{-R}$$

$$\nabla \times \mathbf{H} = 0 \therefore \text{Conservative only}$$